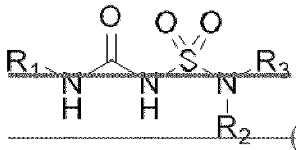
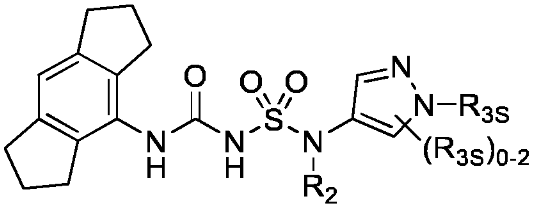
AUXILIARY REQUEST 1

CLAIMS:

1. A compound of Formula (Iac1):



~~(I),~~



(Iac-1),

or a solvate, or pharmaceutically acceptable salt thereof, wherein:

~~**R$_1$** is C_3-C_{16} cycloalkyl optionally substituted by one or more **R$_{1S}$**, wherein the C_3-C_{16} cycloalkyl is a C_3-C_8 monocyclic cycloalkyl or polycyclic cycloalkyl, or, **R$_1$** is a phenyl substituted by two or three **R$_{1S}$**; wherein each **R$_{1S}$** is independently C_1-C_6 alkyl, C_1-C_6 haloalkyl, C_1-C_6 alkoxy, C_1-C_6 haloalkoxy, or halo, and at least one **R$_{1S}$** is C_1-C_6 alkyl, C_1-C_6 haloalkyl, or halo;~~

R$_2$ is –(CH~~X_2X~~$_2$)$_n$-**R$_{2S}$**, wherein n is 0, 1, or 2, and ~~each X_2 is independently H, C_1-C_6 alkyl, C_2-C_6 alkenyl, or C_2-C_6 alkynyl, wherein the C_1-C_6 alkyl, C_2-C_6 alkenyl, or C_2-C_6 alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C_1-C_6 alkyl), -NH_2, -NH(C_1-C_6 alkyl), -N(C_1-C_6 alkyl)$_2$, or oxo;~~

R$_{2S}$ is a 5 or 6~~4- to 8-~~ membered monocyclic heterocycloalkyl optionally substituted with one or more C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, C_1-C_6 haloalkyl, halo, -CN, -OH, -O(C_1-C_6 alkyl), -NH_2, -NH(C_1-C_6 alkyl), -N(C_1-C_6 alkyl)$_2$, or oxo; and

~~**R$_3$** is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more **R$_{3S}$**, wherein~~ each **R$_{3S}$** is independently C_1-C_6 alkyl, C_1-C_6 haloalkyl, C_3-C_8 cycloalkyl, halo, or C_3-C_8 heterocycloalkyl wherein the C_1-C_6 alkyl, C_1-C_6 haloalkyl, C_3-C_8 cycloalkyl, or C_3-C_8 heterocycloalkyl is optionally substituted with –O(C_1-C_6 alkyl), -N(C_1-C_6 alkyl)$_2$, halo, or -CN.

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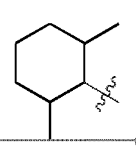
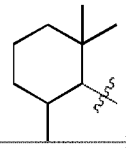
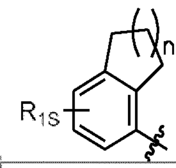
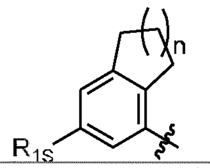
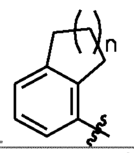
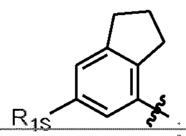
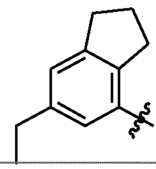
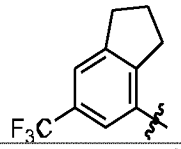
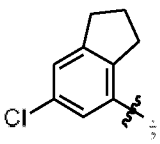
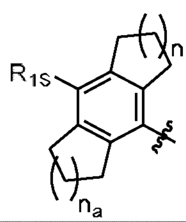
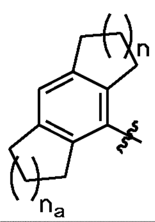
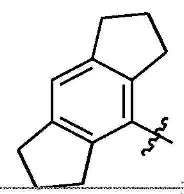
~~2. The compound of claim 1, wherein $\mathbf{R_1}$ is C_3-C_7 monocyclic cycloalkyl, C_9-C_{10} bicyclic cycloalkyl, C_{12}-C_{16} tricyclic cycloalkyl, or phenyl, wherein the C_3-C_7 monocyclic cycloalkyl, C_9-C_{10} bicyclic cycloalkyl, or C_{12}-C_{16} tricyclic cycloalkyl is optionally substituted by one or more $\mathbf{R_{1S}}$, the phenyl is substituted by two or three $\mathbf{R_{1S}}$; and wherein at least one $\mathbf{R_{1S}}$ is C_1-C_6 alkyl, C_1-C_6 haloalkyl, or halo;~~

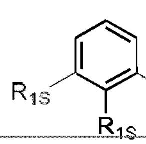
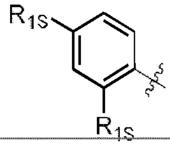
~~optionally wherein:~~

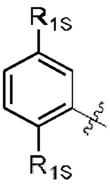
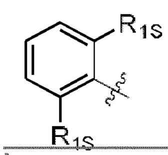
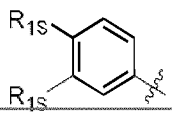
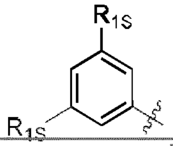
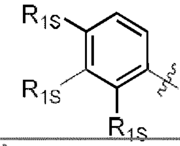
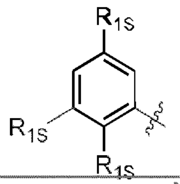
~~(a) the C_3-C_7 monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more $\mathbf{R_{1S}}$; further optionally wherein the C_3-C_7 monocyclic cycloalkyl is~~  ~~or~~ ~~;~~

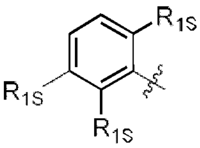
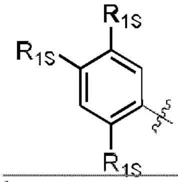
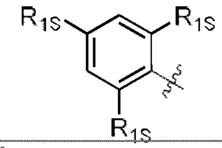
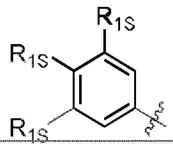
~~(b) the C_9-C_{10} bicyclic cycloalkyl is~~ ~~,~~ ~~, or~~ ~~, wherein n is 0, 1, or 2; optionally~~ ~~; further optionally~~ ~~,~~ ~~, or~~

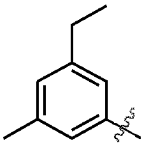
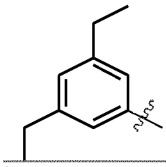
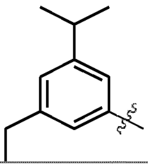
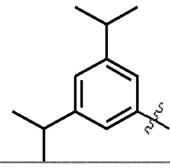
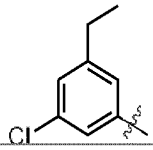
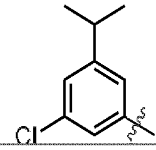
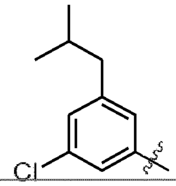
~~;~~

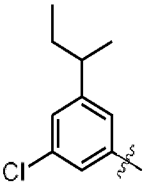
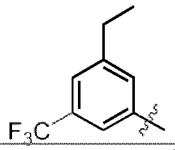
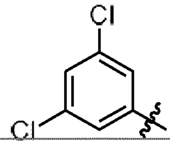
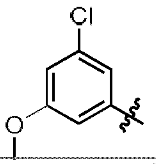
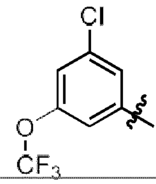
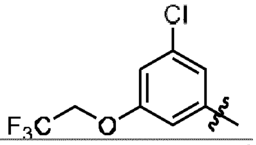
~~(c) the C_{12}-C_{16} tricyclic cycloalkyl is~~  ~~or~~ ~~, wherein n and n_a each independently are 0, 1, 2, or 3; further optionally~~ ~~; and/or~~

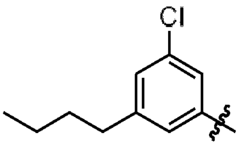
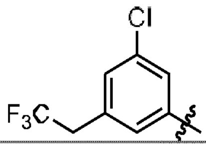
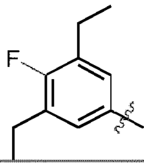
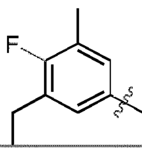
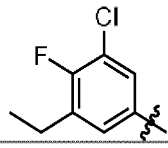
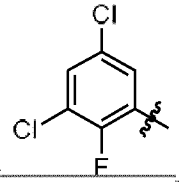
~~(d) the phenyl is substituted by two, or three $\mathbf{R_{1S}}$; optionally~~  ~~,~~  ~~,~~

 ~~,~~  ~~,~~  ~~,~~  ~~,~~  ~~,~~  ~~,~~

 ~~,~~  ~~,~~  ~~, or~~  ~~; or further optionally~~

 ~~,~~  ~~,~~  ~~,~~  ~~,~~  ~~,~~  ~~,~~  ~~,~~

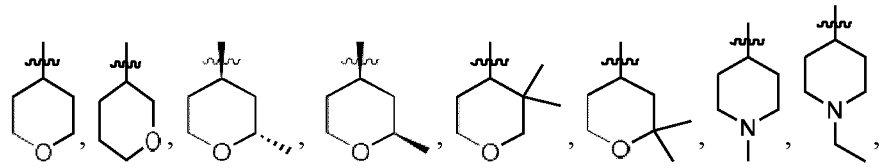
 ~~,~~  ~~,~~  ~~,~~  ~~,~~  ~~,~~  ~~,~~

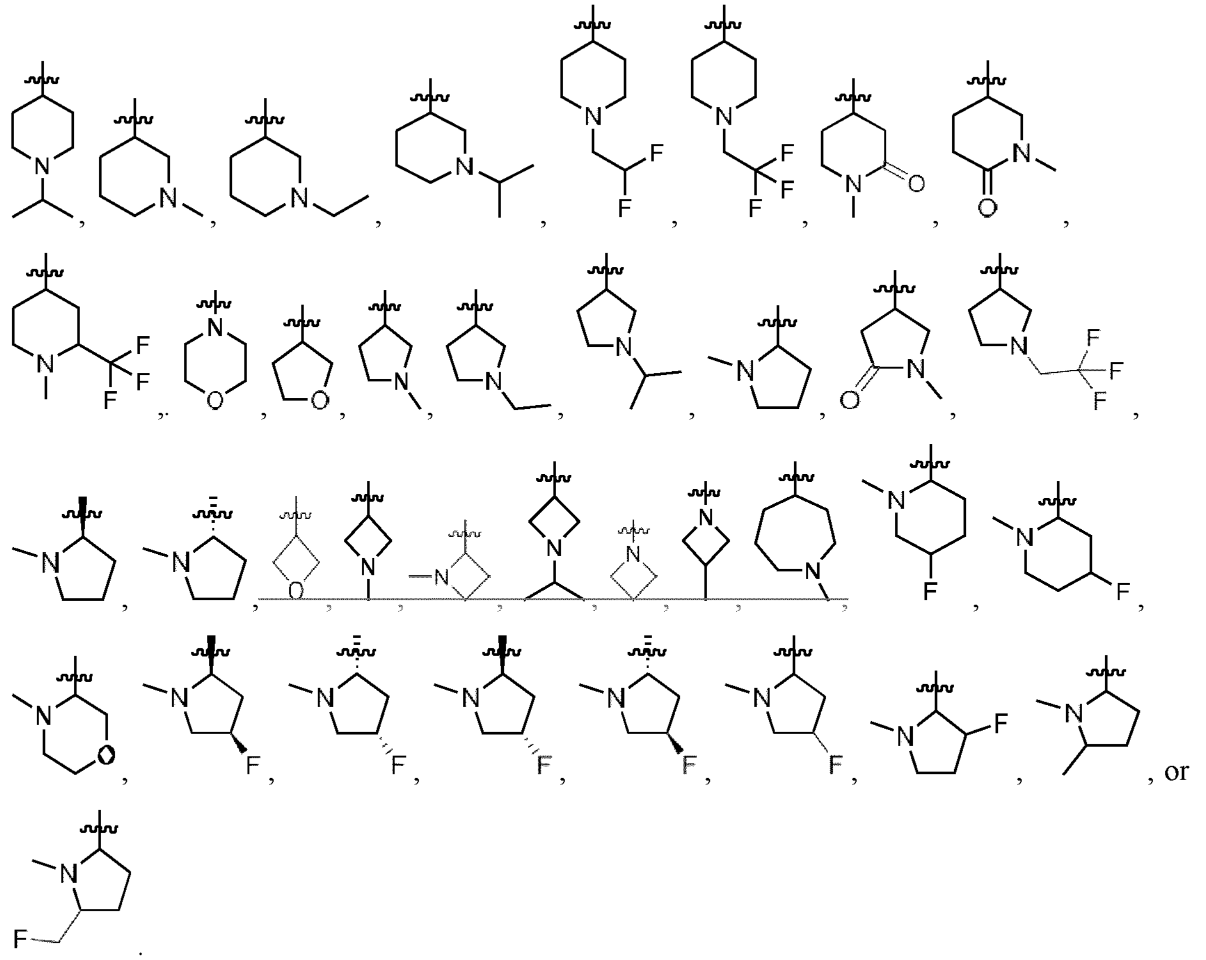
 ~~,~~  ~~,~~   ~~,~~  ~~, or~~  ~~.~~

~~3. The compound of claim 1 or 2, wherein $\mathbf{R_2}$ is $-\mathbf{R_{2S}}$, $-(CX_2X_2)-\mathbf{R_{2S}}$, or $-(CX_2X_2)_2-\mathbf{R_{2S}}$.~~

~~4~~2. The compound of ~~any one of~~ claim~~s~~ 1 ~~to 3~~, wherein $\mathbf{R_{2S}}$ is ~~4- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, C_1-C_6 haloalkyl, halo, -CN, -OH, -O(C_1-C_6 alkyl), -NH_2, -NH(C_1-C_6 alkyl), -N(C_1-C_6 alkyl)$_2$, or oxo;~~

~~Optionally oxetanyl, azetidinyl,~~ tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, or morpholinyl~~, or azepanyl~~, wherein the ~~oxetanyl, azetidinyl,~~ tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, or morpholinyl~~, or azepanyl~~ is optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo; or

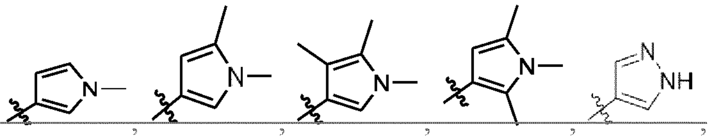
further optionally 

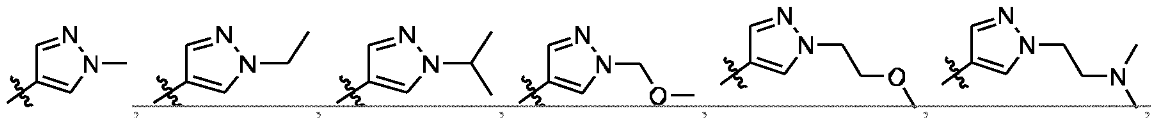


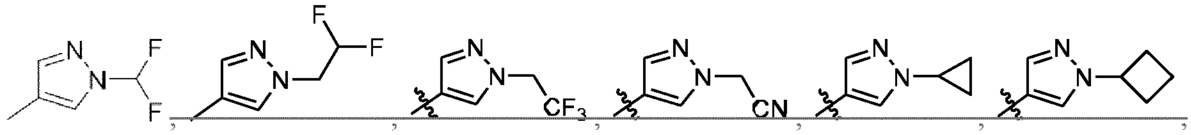
~~5. The compound of any one of claims 1 to 4, wherein R_3 is 7- to 12-membered heterocycloalkyl optionally substituted with one or more R_{3S};~~

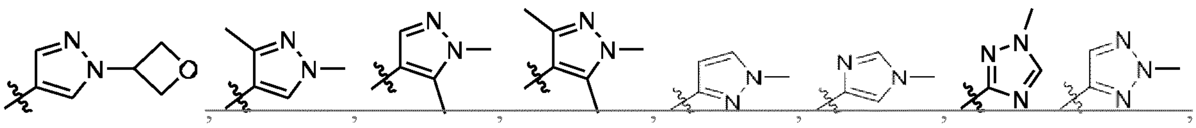
~~optionally wherein R_3 is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-~~

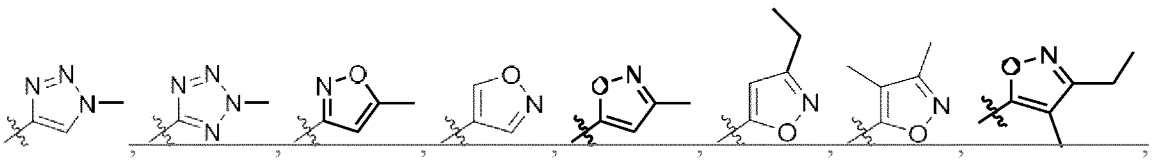
~~tetrahydrobenzo[c]isoxazole, wherein the pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole is optionally substituted with one or more **R₃ₛ**; or~~

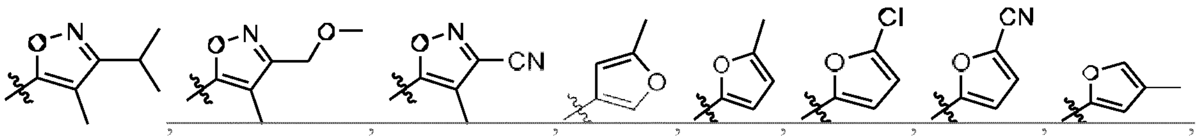
~~further optionally wherein **R₃** is~~ 

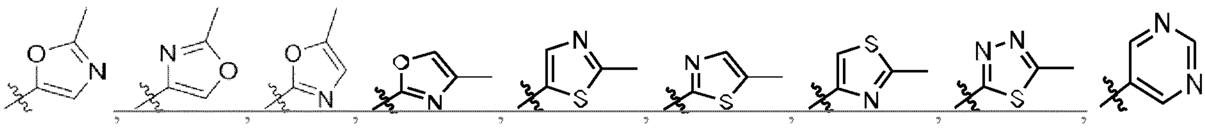


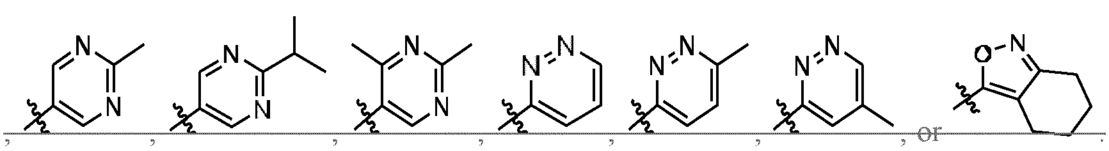




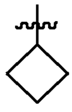
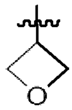




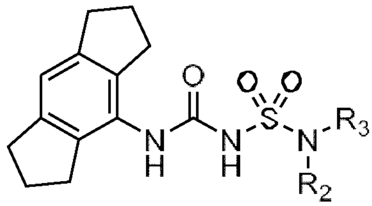




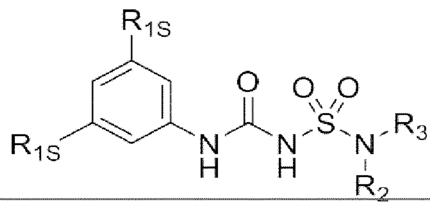
3~~6~~. The compound of ~~any one of~~ claim~~s~~ 1 or claim 2~~to 5~~, wherein at least one **R_{3S}** is C_1-C_6 alkyl, C_1-C_6 haloalkyl, C_3-C_8 cycloalkyl, halo, or C_3-C_8 heterocycloalkyl wherein the C_1-C_6 alkyl, C_1-C_6 haloalkyl, C_3-C_8 cycloalkyl, or C_3-C_8 heterocycloalkyl is optionally substituted with –O(C_1-C_6 alkyl), -N(C_1-C_6 alkyl)$_2$, halo, or -CN;

optionally wherein at least one **R3S** is , , , methyl, ethyl, isopropyl, -CH_2OCH_3, -CH_2CF_3, -$CH_2CH_2OCH_3$, -CH_2CN, CH_2CF_2, or -$CH_2CH_2N(CH_3)_2$.

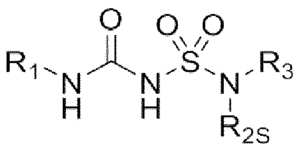
~~7~~4. The compound of any one of the preceding claims, wherein the compound is of Formula ~~(Ia-1), (Ia-2), (Ib-1), (Ib-2), (Ib-3), (Ib-1a), (Ib-1b), (Ib-1c), (Ic-1), (Ic-2), (Ic-3), (Ic-4), (Ic-5), (Ic-6), (Ic-7), (Ic-8), (Iab-1), (Iab-2), (Iac-1), (Iac-2), (Iac-3), (Iac-4), (Ibc-1), (Ibc-2),~~ (Iabc-1), (Iabc-1a), or (Iabc-1b):



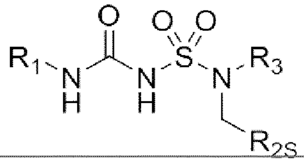
~~(Ia-1);~~



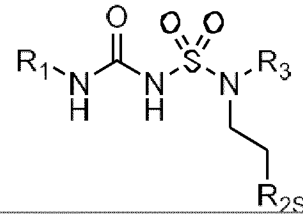
~~(Ia-2);~~



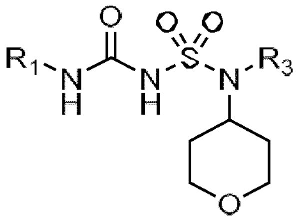
~~(Ib-1);~~



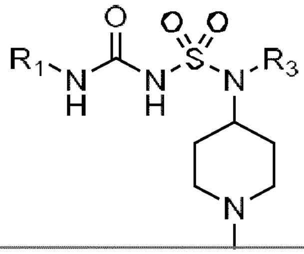
~~(Ib-2);~~



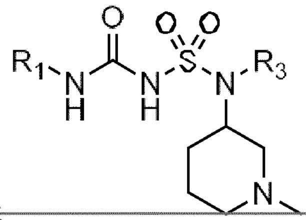
~~(Ib-3);~~



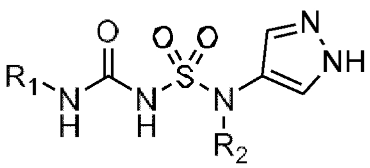
~~(Ib-1a);~~



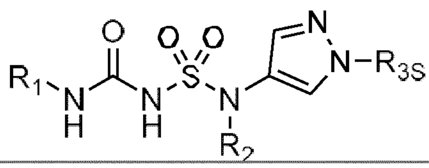
~~(Ib-1b);~~



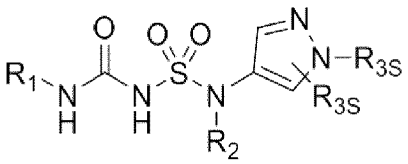
~~(Ib-1c);~~



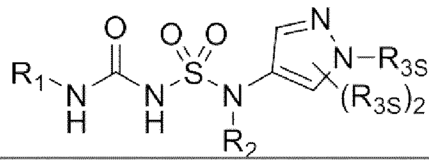
~~(Ic-1);~~



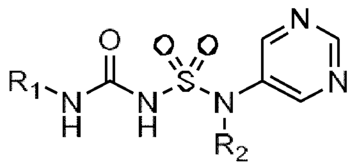
~~(Ic-2);~~



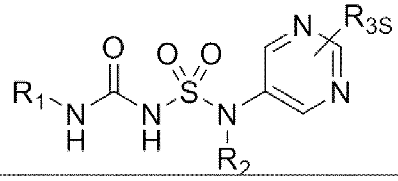
~~(Ic-3);~~



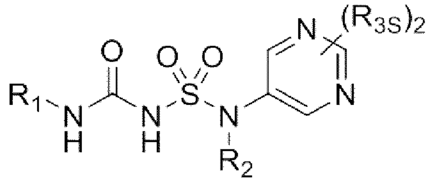
~~(Ic-4);~~



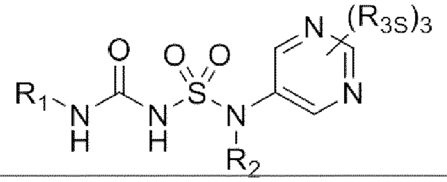
~~(Ic-5);~~



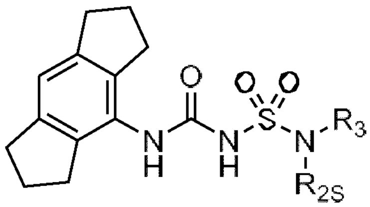
~~(Ic-6);~~



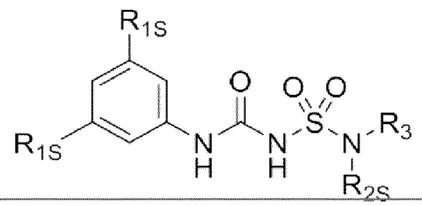
~~(Ic-7);~~



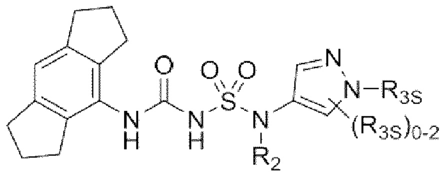
~~(Ic-8),~~



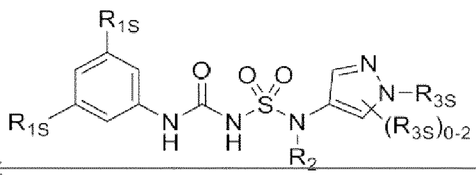
~~(Iab-1);~~



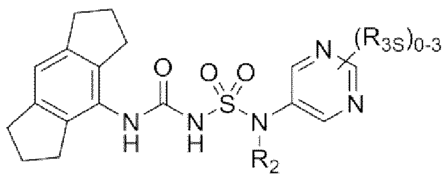
~~(Iab-2);~~



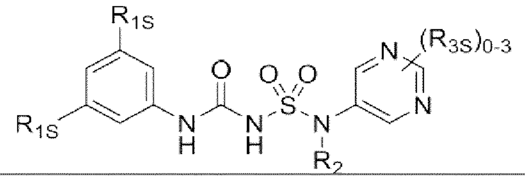
~~(Iac-1) ;~~



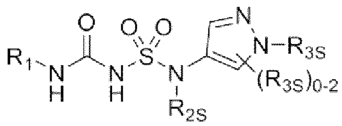
~~(Iac-2);~~



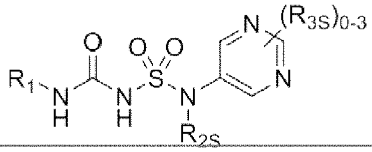
~~(Iac-3);~~



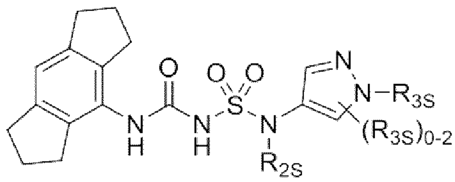
~~(Iac-4),~~



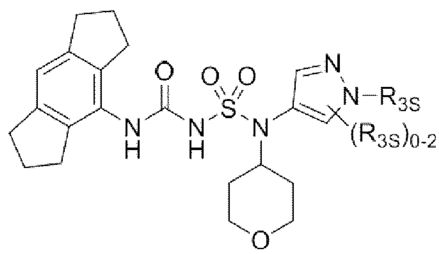
~~(Ibc-1);~~



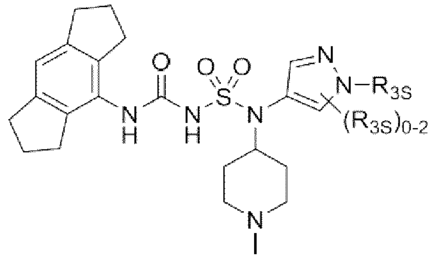
~~(Ibc-2),~~



(Iabc-1);



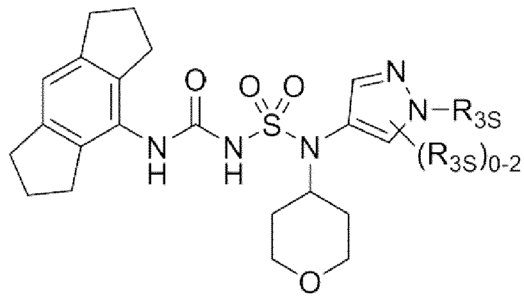
(Iabc-1a); or



(Iabc-1b)

or a solvate, or pharmaceutically acceptable salt thereof, wherein ~~R_1,~~ $\mathbf{R_{2S}}$, ~~R_2,~~ and $\mathbf{R_{3S}}$ are as described ~~herein~~in claim 1.

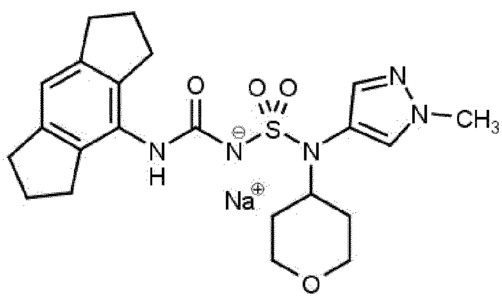
~~8~~5. The compound of any one of claims 1 to ~~7~~4, wherein the compound is of Formula (Iabc-1a):

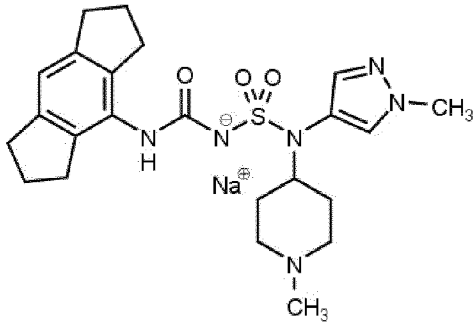
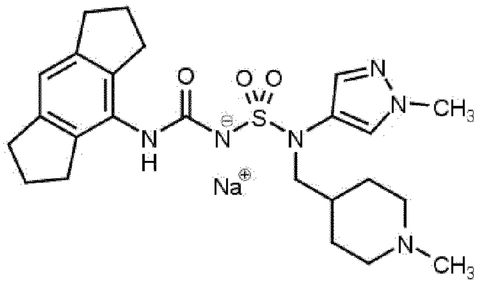
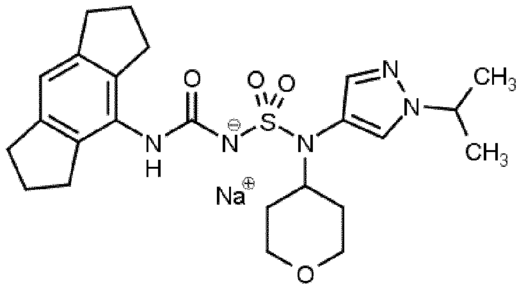
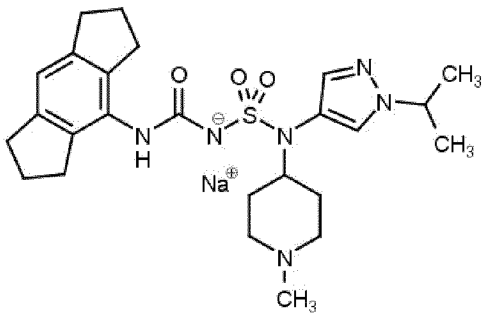


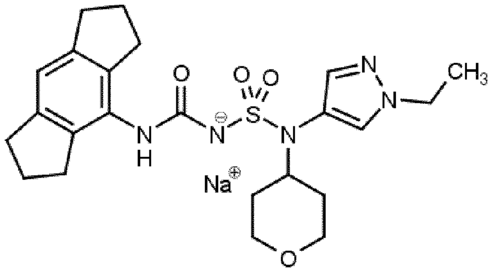
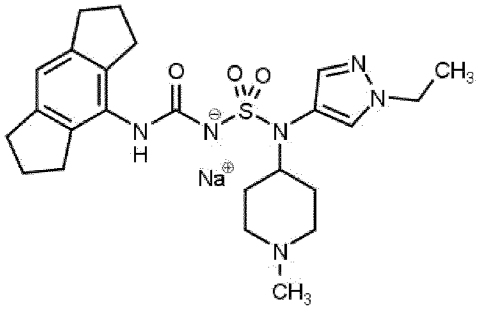
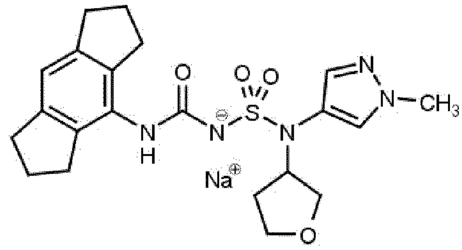
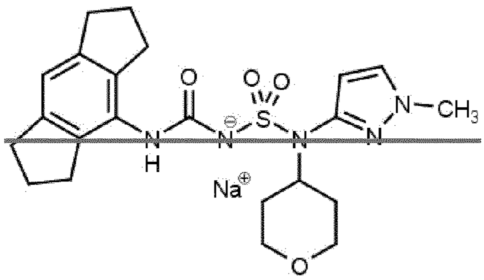
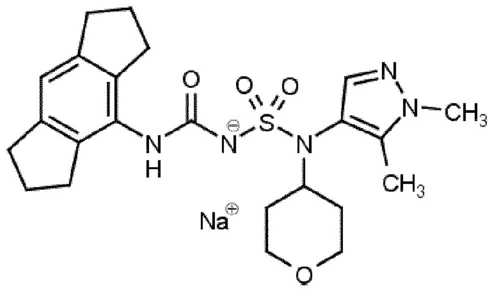
(Iabc-1a)

or a solvate, or pharmaceutically acceptable salt thereof, wherein $\mathbf{R_{3S}}$ is as described ~~herein~~in claim 1.

~~9~~6. The compound of any one of ~~the~~ claims 1 to ~~8~~5, being selected from:

Compound No.	**Structure**
1	

~~Compound No.~~	~~Structure~~
2	
3	
4	
5	

~~**Compound No.**~~	~~**Structure**~~
6	
7	
8	
~~**9**~~	
10	

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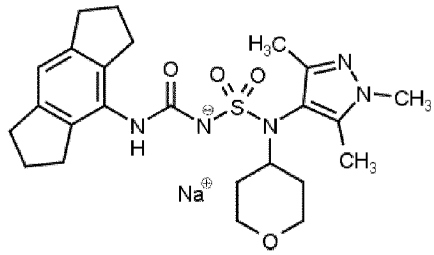
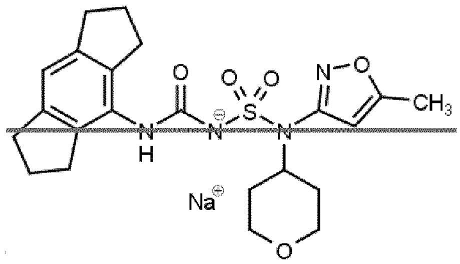
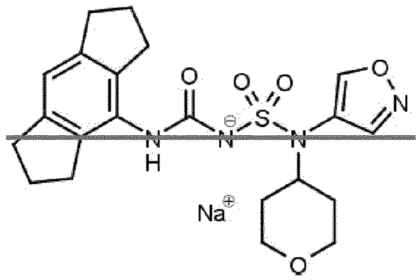
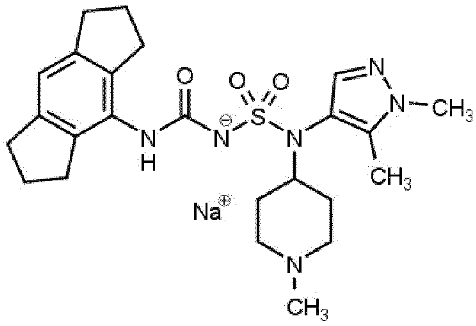
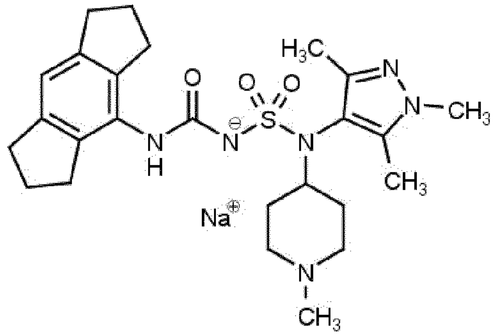
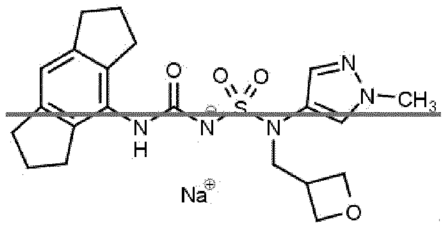
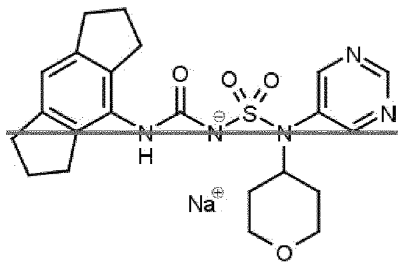
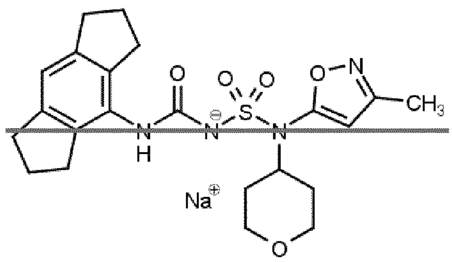
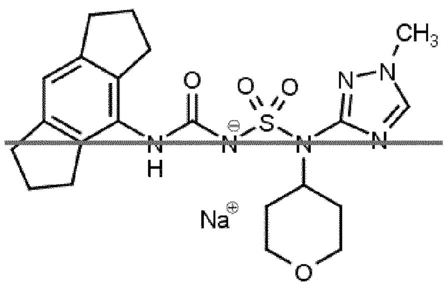
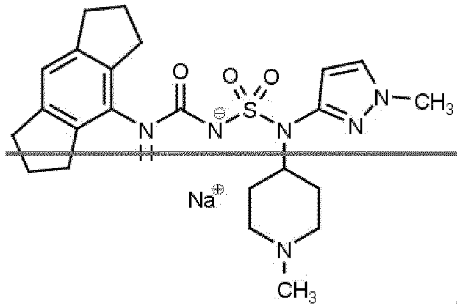
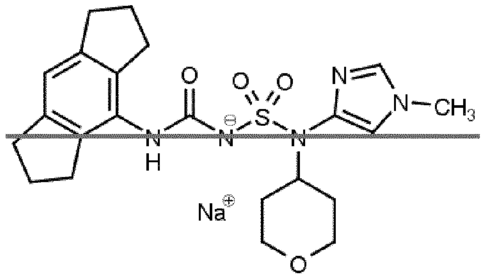
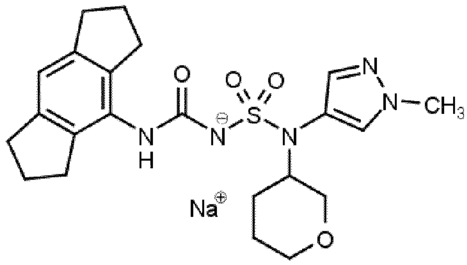
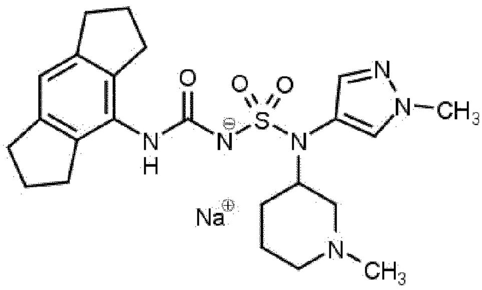
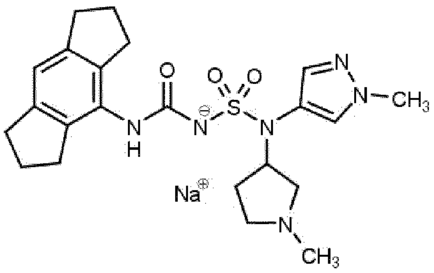
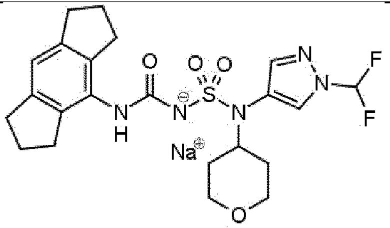
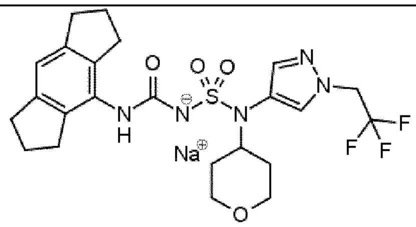
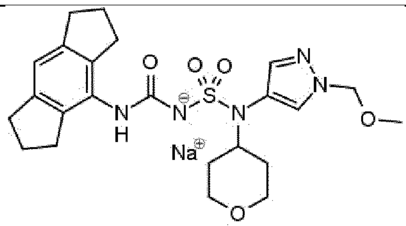
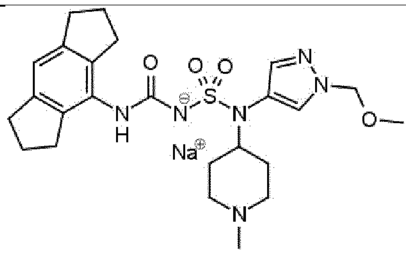
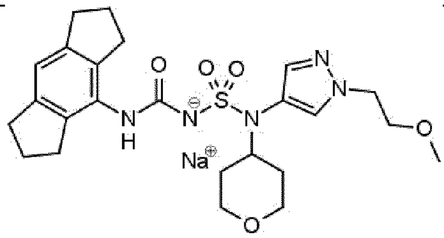
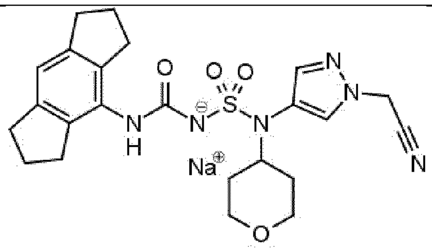
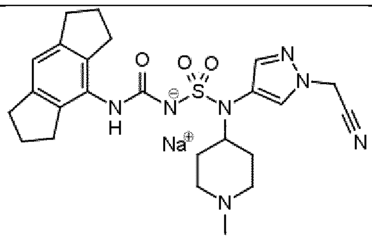
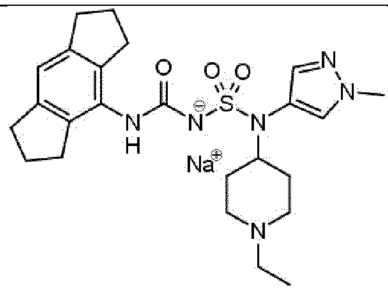
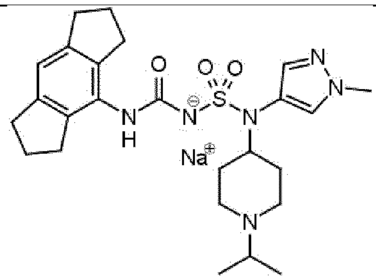
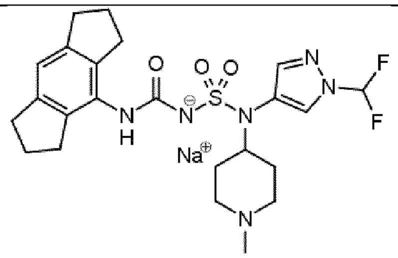
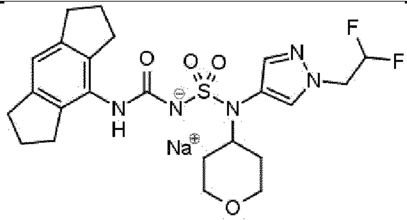
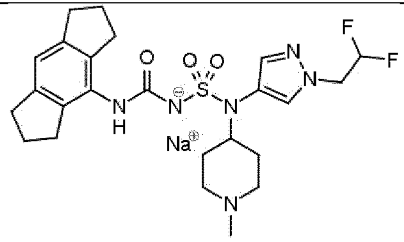
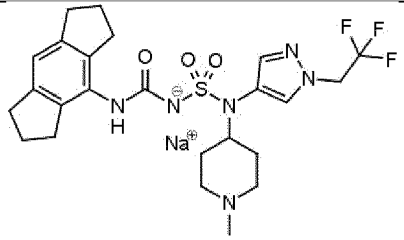
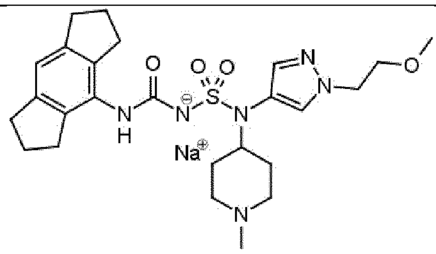
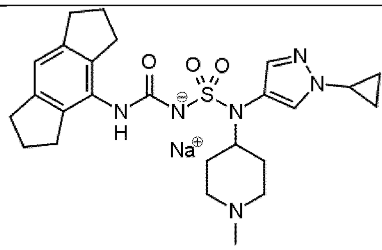
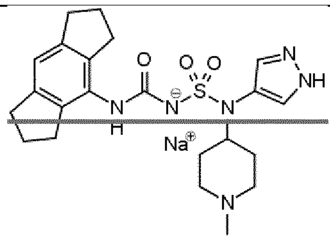
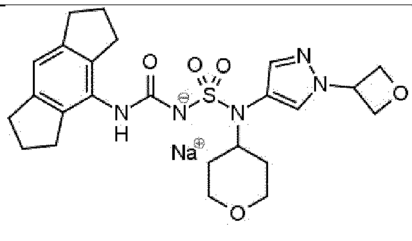
~~Compound No.~~	~~Structure~~
11	
~~12~~	
~~13~~	
14	

~~Compound No.~~	~~Structure~~
15	
~~16~~	
~~17~~	
~~18~~	
~~19~~	

~~Compound No.~~	~~Structure~~
~~20~~	
~~21~~	
22	
23	

~~Compound No.~~	~~Structure~~
24	
25	
26	
27	
28	
29	

~~Compound No.~~	~~Structure~~
30	
31	
32	
33	
34	
35	

~~Compound No.~~	~~Structure~~
36	
37	
38	
39	
~~40~~	
41	

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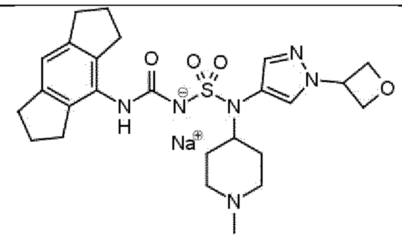
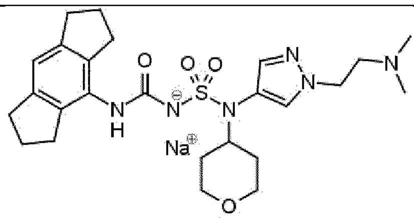
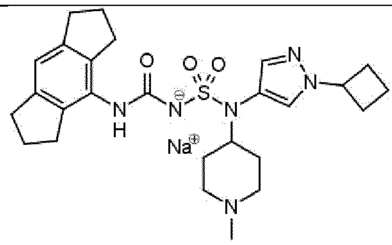
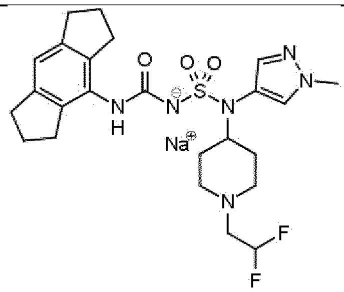
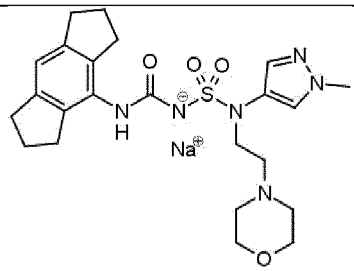
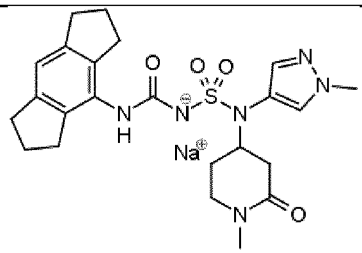
42	
43	
44	
45	
46	
47	

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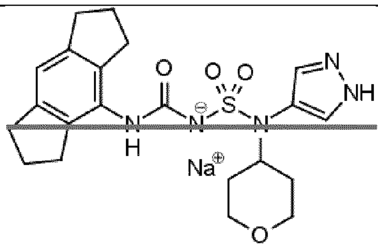
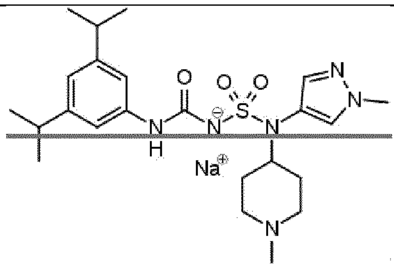
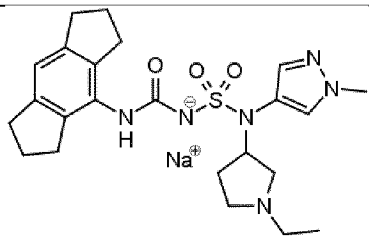
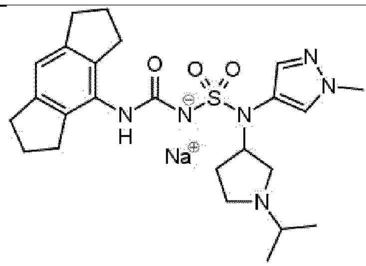
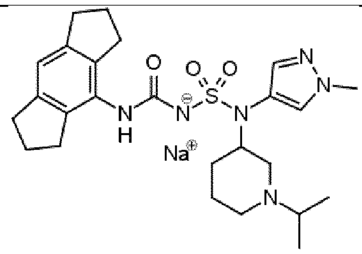
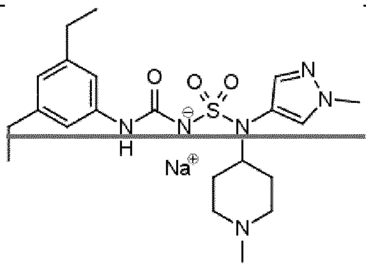
~~48~~	
~~49~~	
50	
51	
52	
~~53~~	

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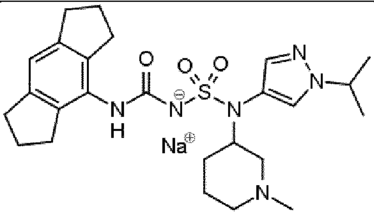
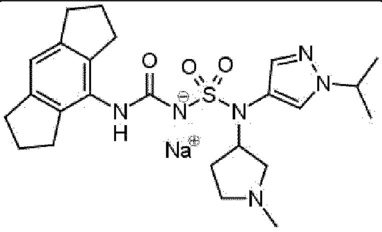
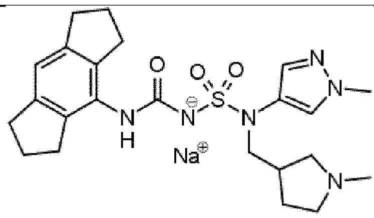
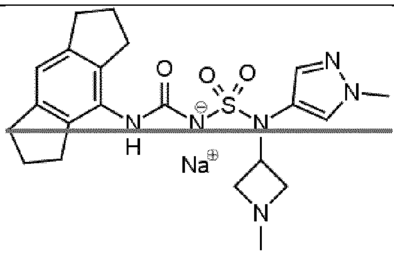
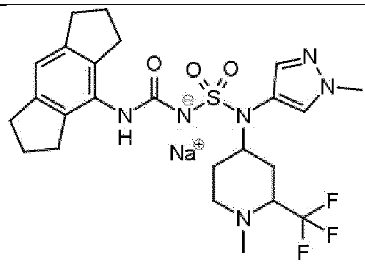
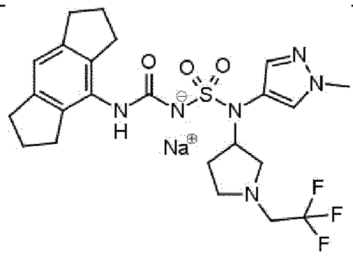
55	
56	
57	
~~59~~	
60	
61	

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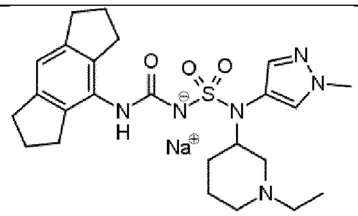
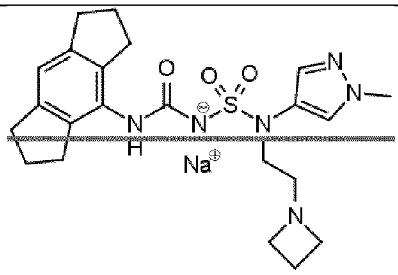
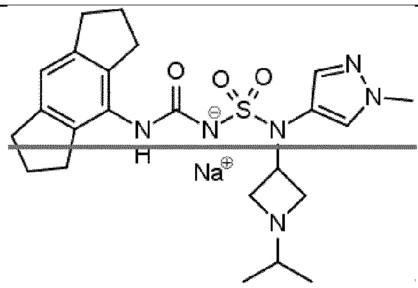
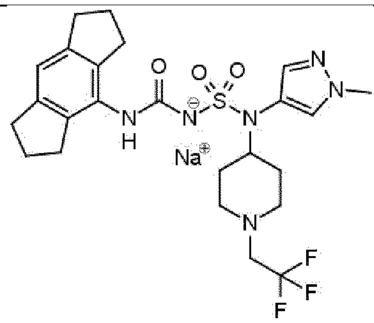
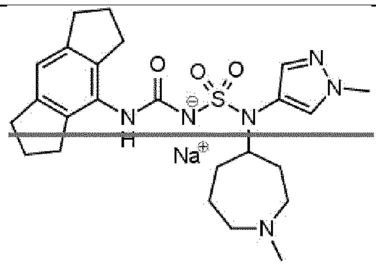
62	
~~63~~	
~~64~~	
65	
~~67~~	

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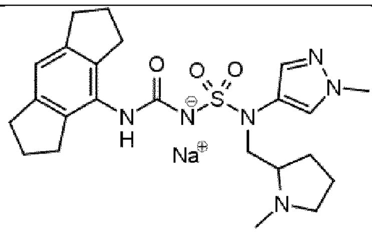
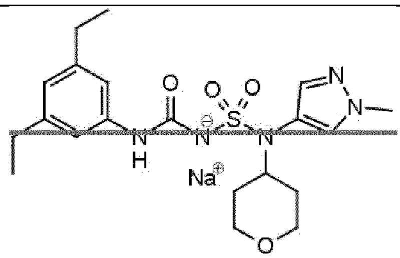
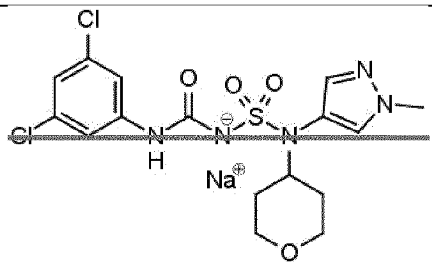
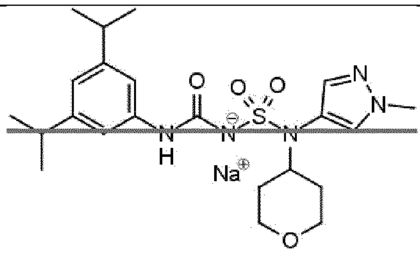
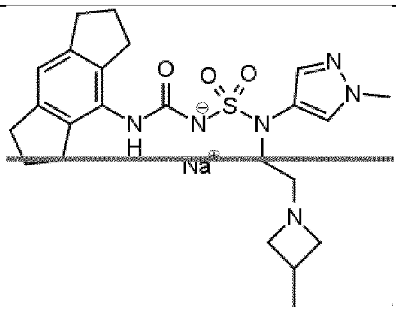
68	
~~70~~	
~~72~~	
~~73~~	
~~74~~	

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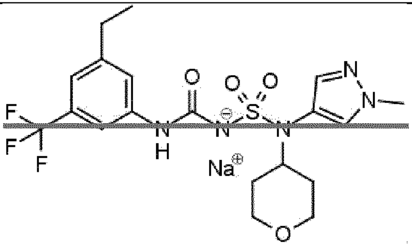
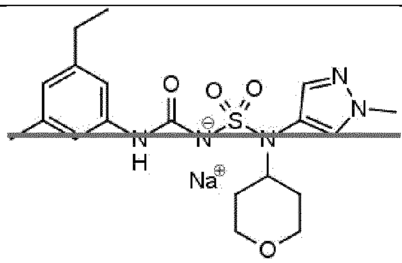
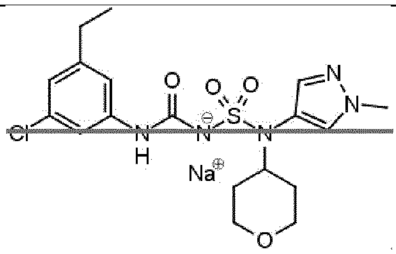
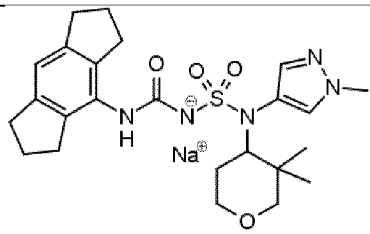
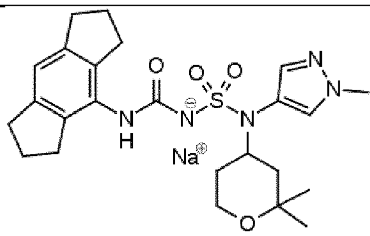
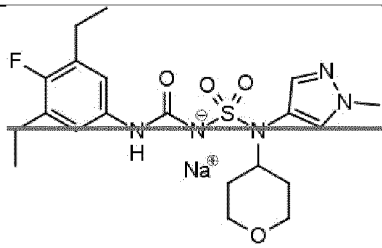
~~75~~	
~~76~~	
~~77~~	
78	
79	
~~80~~	

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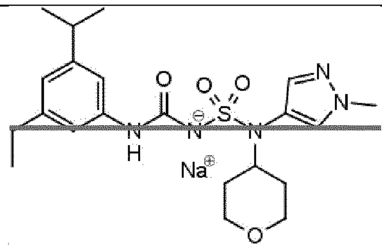
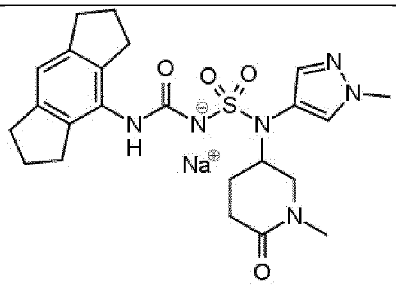
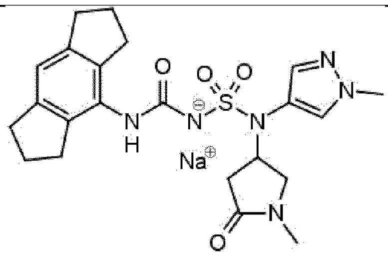
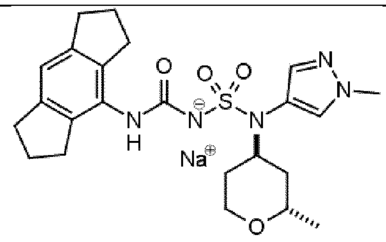
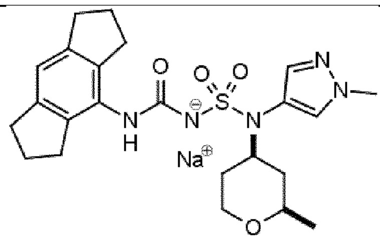
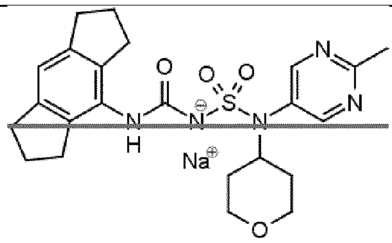
~~81~~	
82	
83	
84	
85	
~~86~~	

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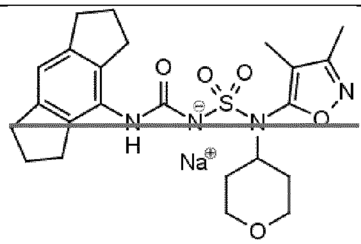
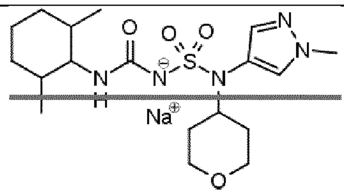
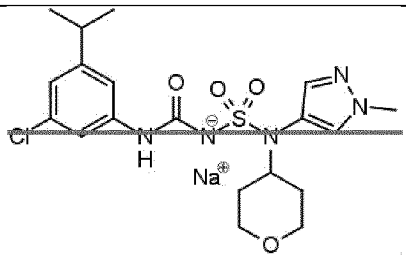
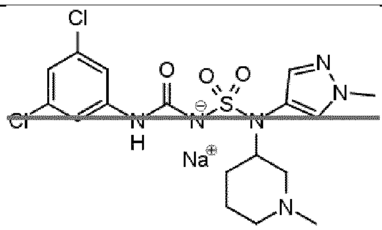
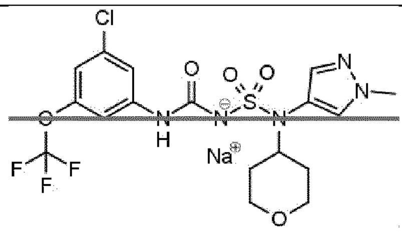
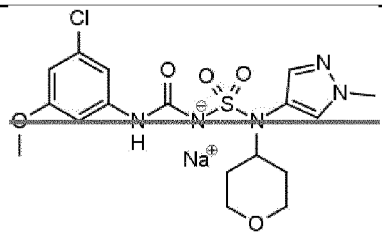
~~87~~	
~~88~~	
~~89~~	
~~91~~	
~~92~~	
~~93~~	

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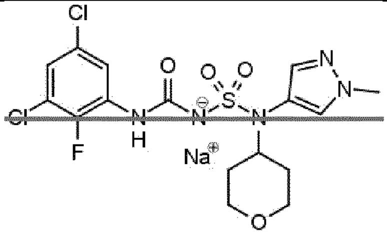
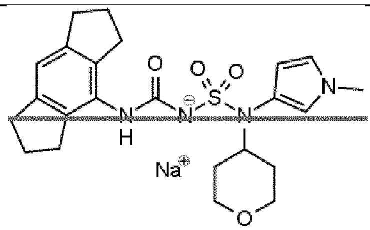
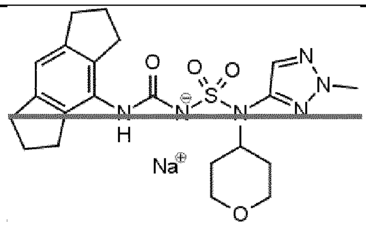
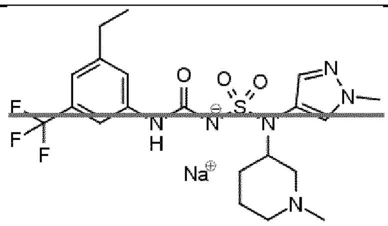
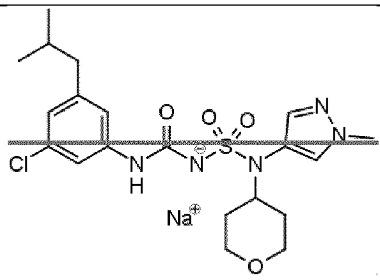
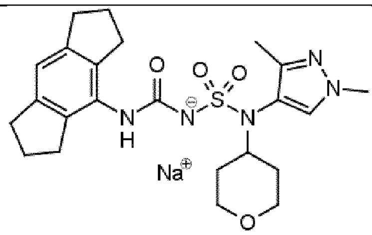
~~94~~	
~~95~~	
~~96~~	
~~97~~	
~~98~~	
99	

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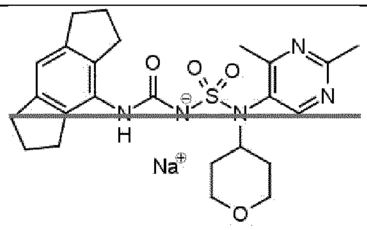
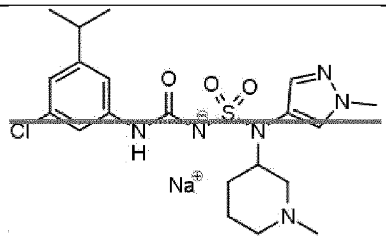
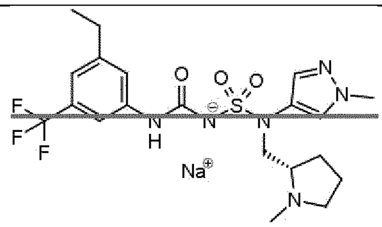
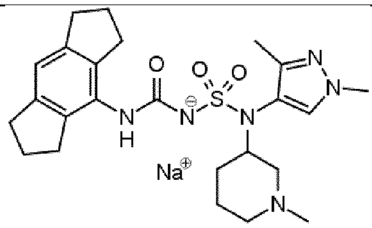
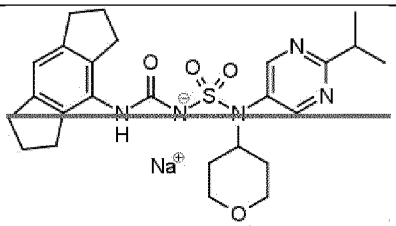
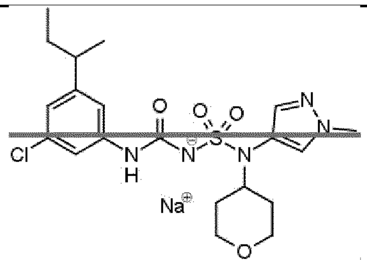
~~100~~	
~~101~~	
~~102~~	
103	
~~104~~	
~~105~~	

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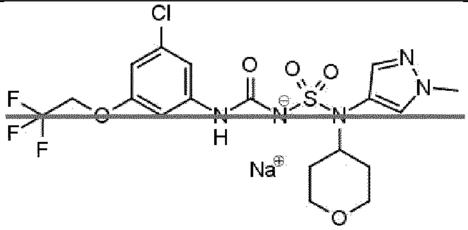
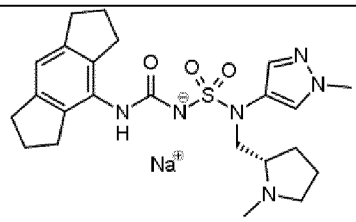
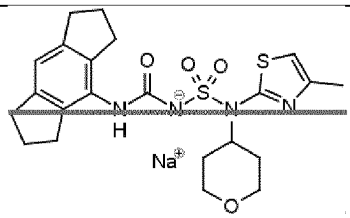
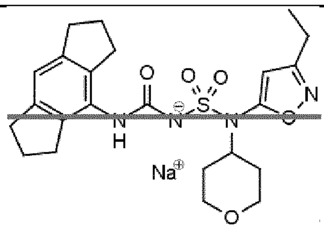
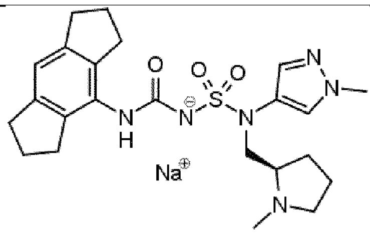
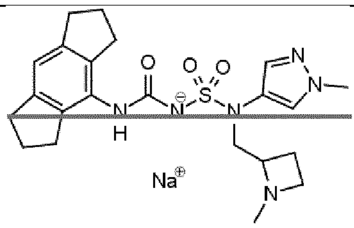
~~106~~	
107	
~~108~~	
~~109~~	
110	
~~111~~	

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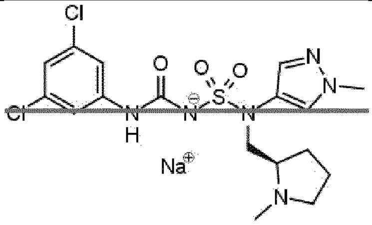
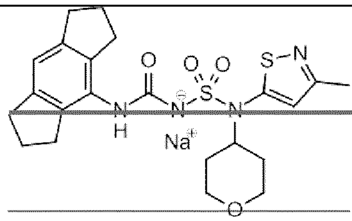
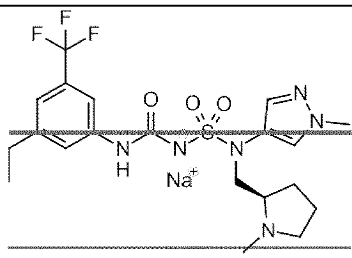
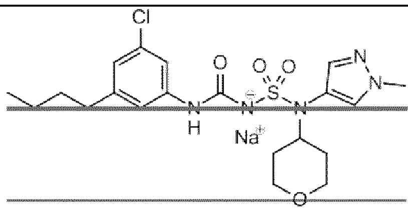
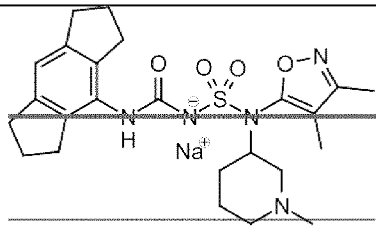
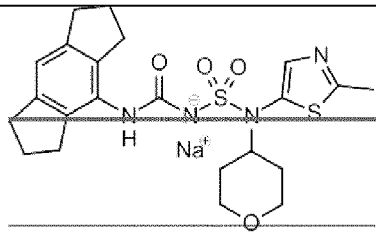
~~112~~	
~~113~~	
~~114~~	
~~115~~	
~~116~~	
~~117~~	

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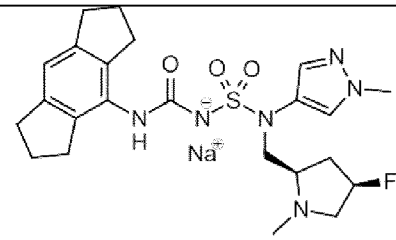
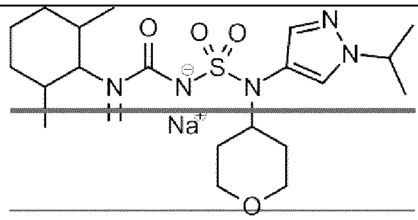
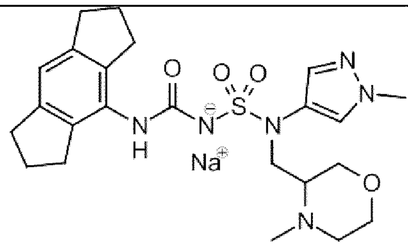
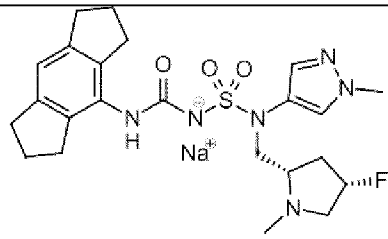
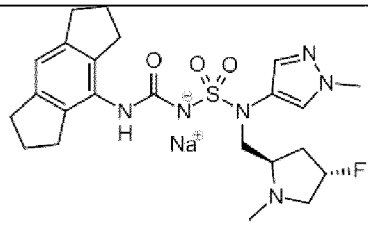
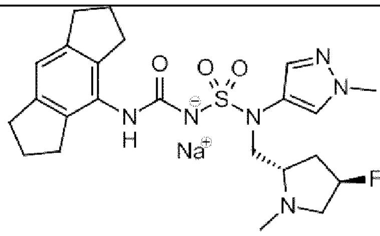
118	
~~119~~	
120	
121	
122	
123	

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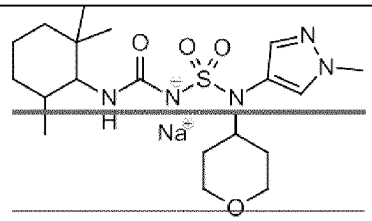
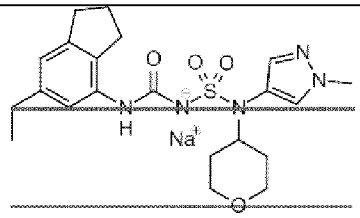
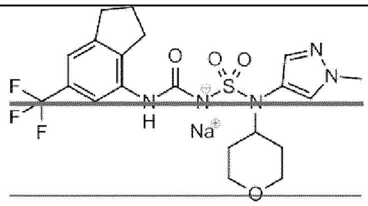
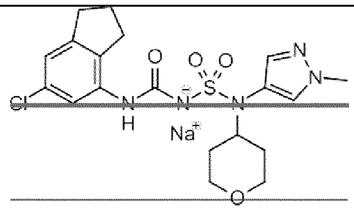
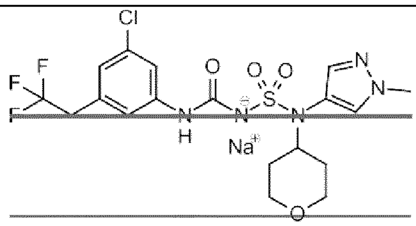
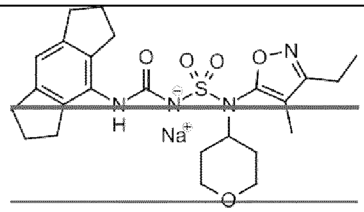
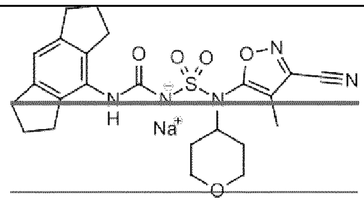
~~124~~	
~~125~~	
~~126~~	
~~127~~	
~~128~~	
~~129~~	
~~130~~	

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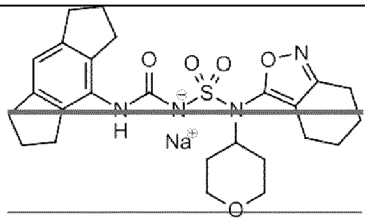
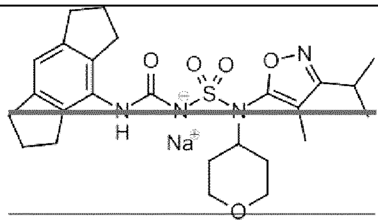
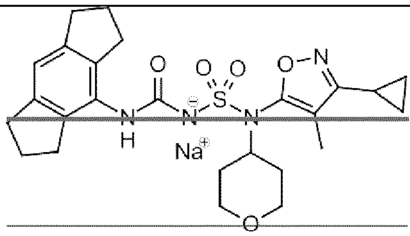
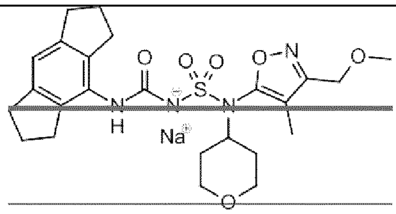
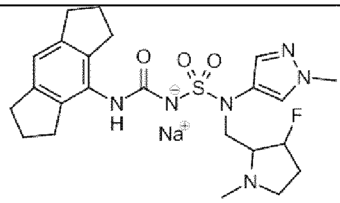
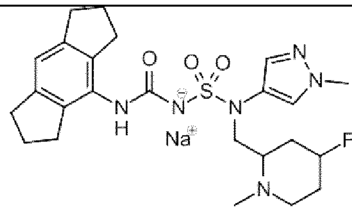
~~131~~	
~~132~~	
~~133~~	
~~134~~	
135	
136	

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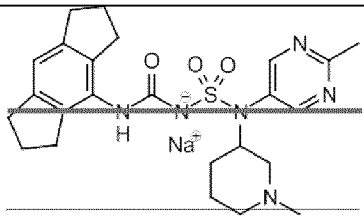
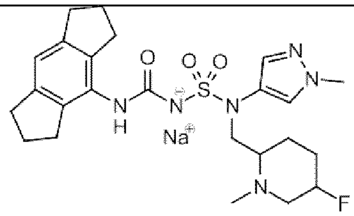
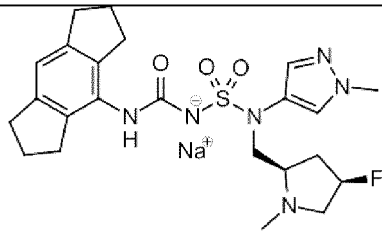
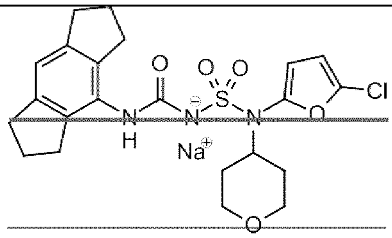
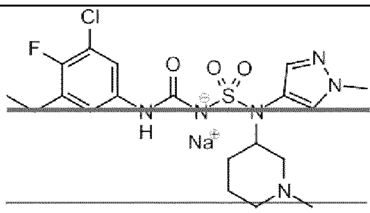
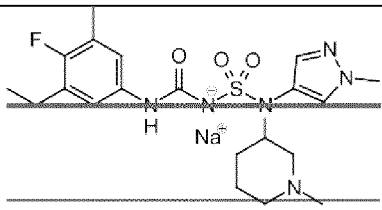
~~137~~	
138	
139	
~~140~~	
~~141~~	
~~142~~	

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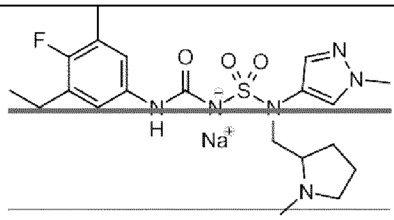
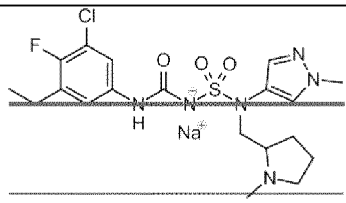
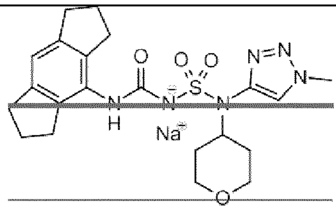
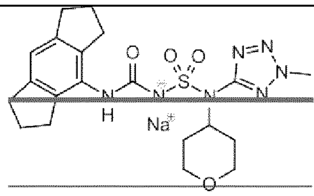
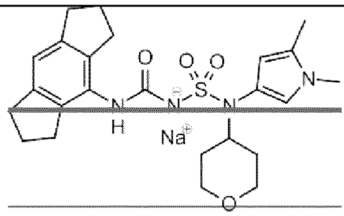
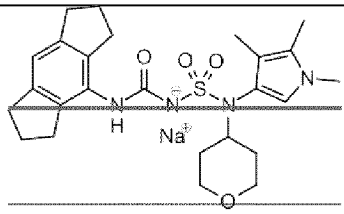
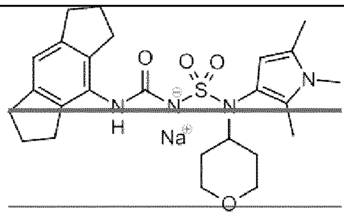
~~143~~	
~~144~~	
~~145~~	
~~146~~	
~~147~~	
~~148~~	
~~149~~	

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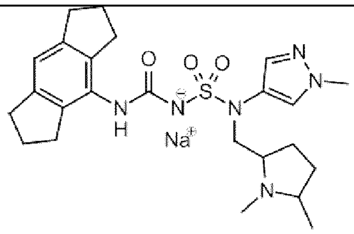
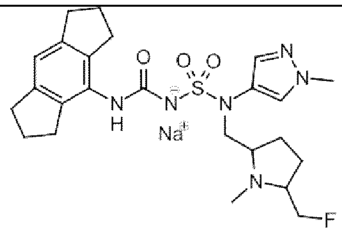
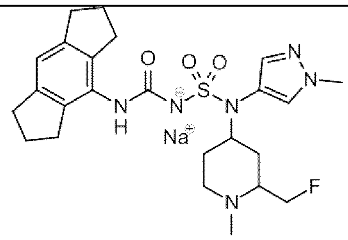
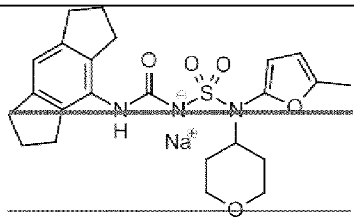
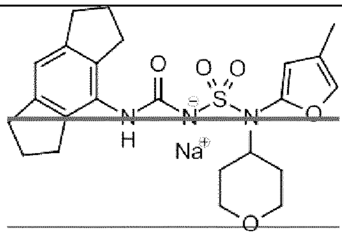
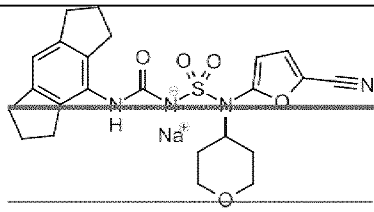
150	
151	
152	
~~153~~	
~~154~~	
~~155~~	

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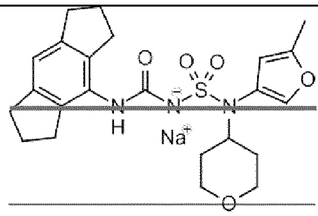
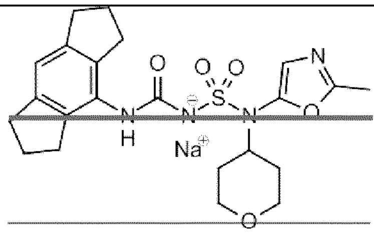
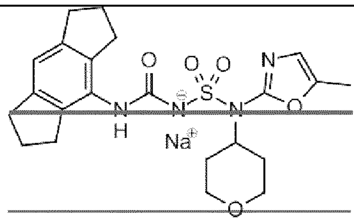
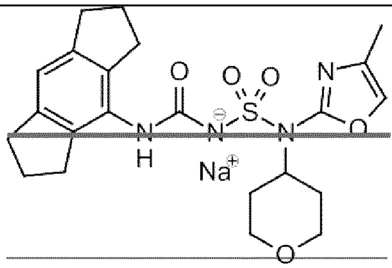
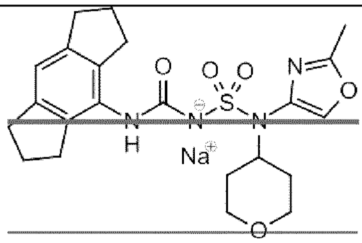
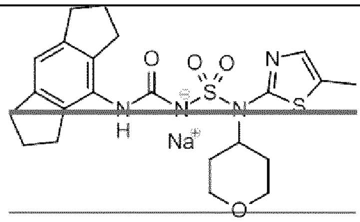
~~156~~	
~~157~~	
~~158~~	
~~159~~	
~~160~~	
~~161~~	

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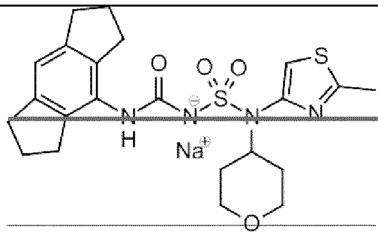
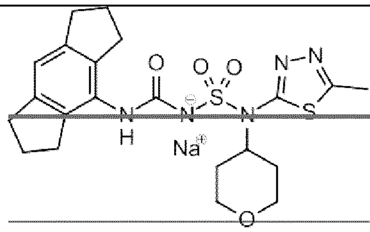
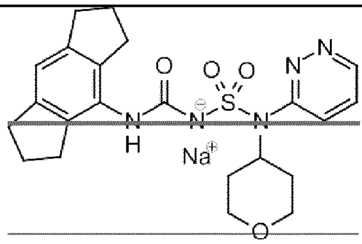
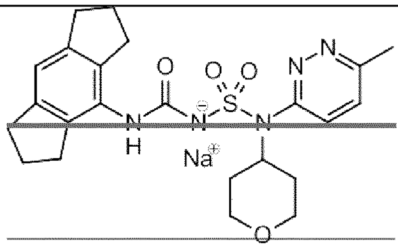
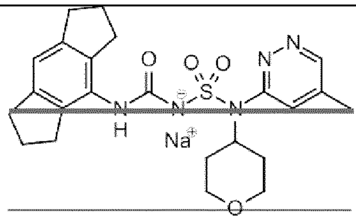
~~162~~	
~~163~~	
~~164~~	
~~165~~	
~~166~~	

and pharmaceutically acceptable salts thereof.

~~10~~7. A pharmaceutical composition comprising the compound of any one of claims 1 to ~~9~~ 6 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.

~~11~~8. The compound of any one of claims 1 to ~~9~~6, or the pharmaceutical composition of claim ~~10~~7, for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

~~12~~9. The compound of any one of claims 1 to ~~9~~6, or the pharmaceutical composition of claim ~~10~~7, for use in treating or preventing a disease or disorder; preferably wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

~~13~~10. The compound or pharmaceutical composition for use of claim ~~11~~ 8 or claim ~~12~~9, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

~~14~~11. The compound or pharmaceutical composition for use of claim~~s~~ ~~11~~ 8 or claim ~~12~~9, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

~~15~~12. The compound or pharmaceutical composition for use of claim~~s~~ ~~11~~ 8 or claim ~~12~~9, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.